FireWall

What is firewall ?

- A **firewall** is a **security device/software** that controls **incoming and outgoing network traffic**.
- It works like a **security guard** between **trusted network (internal)** and **untrusted network (internet)**
- It **allows safe traffic** and **blocks harmful or unauthorized traffic**.

Purpose of a Firewall

- **Protect the network** from cyber attacks.
- **Block unauthorized access**.
- **Allow only approved communication**.
- **Monitor network traffic**.
- **Prevent malware, exploits, and intrusions**.
- **Create security boundaries** between networks.

How Does a Firewall Work?

- You give the firewall a set of **rules**.
- Every incoming/outgoing data packet must pass through the **firewall first**.

- The firewall:

    - ✔️ *Allows* traffic if it matches a safe rule
    - ❌ *Blocks* traffic if it looks harmful or unauthorized

- Modern firewalls do much more than simple rule checking — they also detect:

    - Malware
    - Intrusions
    - Suspicious patterns
    - Dangerous websites

Types of Firewall

- Packet Filtering Firewall.
- Stateful Inspection Firewall.
- Proxy Firewall (Application-Level)
- Next-Generation Firewall (NGFW)
- WAF (Web Application Firewall)

Stateless Firewall (Layer 3 & Layer 4)

- Works only using **fixed rules** (ACLs).
- Does **NOT remember past traffic**.
- Treats every packet as **new**, even if it is part of an old connection.

How it works

- Compares each packet with rule list.
- No session tracking.
- Fast but less intelligent.

Limitations

- Cannot apply **advanced/relationship-based** policies.
- Cannot identify if a packet belongs to a good or bad connection.

Example

- If 3 packets from an attacker are blocked, the firewall still checks the **4th packet again** instead of automatically blocking it.

Stateful Firewall (Layer 3 & Layer 4)

- Works using rules + remembers past connections.
- Maintains a **state table** (connection history).
- More secure than stateless firewalls.

How it works

- If a connection is **accepted once**, future packets from that connection are **auto-allowed**.
- If a connection is **denied once,** future packets from that source are **auto-denied**.

Advantages

- Smarter decisions
- Recognizes legit sessions
- Blocks malicious repeated attempts

Proxy Firewall (Application-Level Gateway) Layer 7

- Works as a **middleman** between user and the internet.
- Can inspect the **actual content** inside packets.

What it can do

- Understand HTTP, FTP, DNS, etc.
- Provide **content filtering** (block websites, keywords).
- Hide internal IPs → gives anonymity.
- Inspect data deeply (like URLs, requests, payloads).

Best for

- Web browsing control
- Web filtering
- Extra secure environments

Next-Generation Firewall (NGFW) – Layer 3 to Layer 7

Most Advanced Firewall

NGFW combines all previous firewall features + many security protections.

Key Features

- **Deep Packet Inspection (DPI)**
- **Intrusion Prevention System (IPS)**
- **Heuristic Analysis** (detects attack patterns)
- **SSL/TLS Decryption** (inspects encrypted traffic)
- **Application Control** (identify Facebook, YouTube, SSH, etc.)
- **Threat Intelligence Integration**

Why it's the best

- Blocks attacks in **real-time**
- Detects anomalies
- Understands applications
- Most suitable for modern networks

Comparison Table

Firewall Type	Characteristics
Stateless Firewall	- Basic filtering - No memory of past connections - Very fast, good for high-speed networks
Stateful Firewall	- Understands traffic patterns - Maintains connection history - Can apply complex rules
Proxy Firewall	- Inspects packet contents - Supports content filtering - Application control - Masks internal IPs
NGFW (Next-Generation)	- Advanced threat protection - Built-in IPS - Heuristic/anomaly detection - Decrypts & inspects SSL/TLS traffic

Firewall Rules

- A firewall gives full **control over your network traffic**.
- Even though firewalls come with **built-in default rules**, organizations can create **custom rules** based on their needs.

Example:

- One network may **block all SSH traffic**.
- Another network may **allow SSH only from specific IPs**.

These custom behaviors are configured using **firewall rules**.

Basic Components of a Firewall Rule

Every firewall rule has the following parts:

1.Source Address

- The **IP address** from where the traffic is coming.

2.Destination Address

- The **IP address** of the system receiving the traffic.

3.Port

- The port number used by the traffic (e.g., 80 for HTTP, 22 for SSH).

4.Protocol

- The communication protocol being used
  Example: **TCP, UDP, ICMP**

5. Action

- What the firewall should do when traffic matches this rule.
  (**Allow / Deny / Forward**)

6. Direction

- Whether the rule applies to **incoming (Inbound)** or **outgoing (Outbound)** traffic.

Types of Actions in Firewall Rules

✔️ 1. Allow

- Traffic that matches the rule will be **permitted**.

Example Rule: Allow outbound HTTP traffic

Action	Source	Destination	Protocol	Port	Direction
Allow	192.168.1.0/24	Any	TCP	80	Outbound

❌ 2. Deny

- Traffic matching the rule will be **blocked**.

Used to:

- Block malicious IP addresses
- Restrict unsafe ports
- Reduce attack surface

Example Rule: Deny inbound SSH traffic

Action	Source	Destination	Protocol	Port	Direction
Deny	Any	192.168.1.0/24	TCP	22	Inbound

🔁 3. Forward

- Redirects traffic to another device or network segment.
- Used in firewalls that also act as **routers/gateways**.

Example Rule: Forward inbound HTTP to internal web server

Action	Source	Destination	Protocol	Port	Direction
Forward	Any	192.168.1.8	TCP	80	Inbound

Directionality of Firewall Rules

Rules are categorized based on the **direction** of the traffic they control.

1.Inbound Rules

- Apply to **traffic coming into the network**.
- **Example:**
  ✔️ Allow incoming HTTP (port 80) to your web server.

2.Outbound Rules

- Apply to **traffic going out of the network**.
- **Example:**
  ❌ Block outgoing SMTP (port 25) from all devices except your mail server.

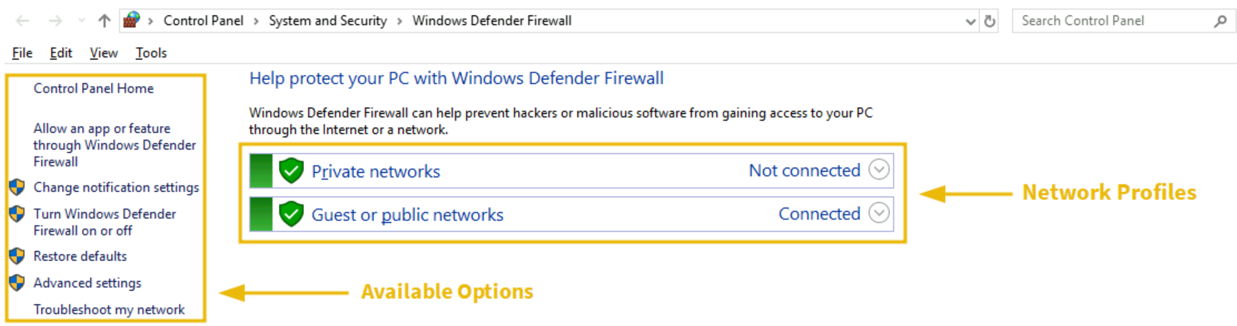
3. Forward Rules

- Used to redirect traffic **inside the network**.
- **Example:**
  ✔️ Forward inbound HTTP (80) to internal web server *192.168.1.8*.

Firewall rule = Source + Destination + Port + Protocol + Action + Direction

- **Actions =** Allow, Deny, Forward
- **Inbound =** controls incoming traffic
- **Outbound =** controls outgoing traffic
- **Forward =** redirect traffic inside network

<u>Windows Defender Firewall (Windows)</u>



Windows Defender Firewall is the **built-in firewall** in Windows OS.

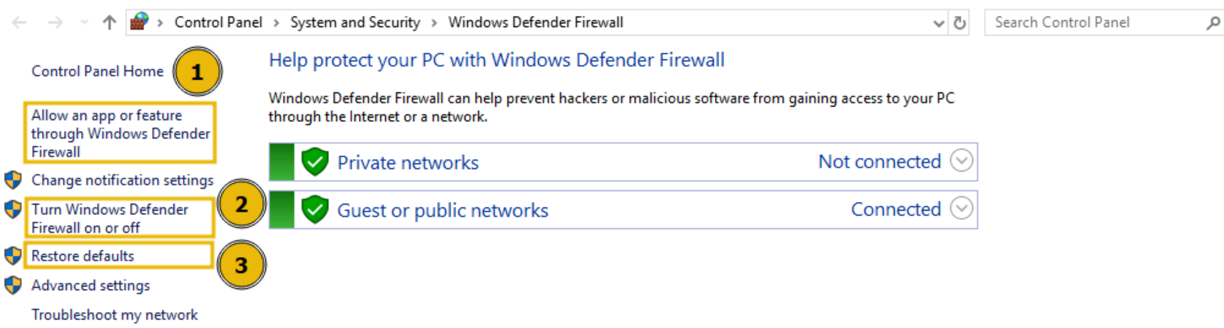
It allows you to:

- Allow/block apps
- Create custom rules
- Control incoming/outgoing network traffic

To open it:

1. Click **Windows Search**
2. Type **Windows Defender Firewall**
3. Open it

Windows Defender Firewall Dashboard



The main dashboard shows:

- Network Profiles
- Settings options
- Allow/Block apps
- Turn firewall on/off
- Restore Defaults

🌐 Network Profiles

Windows uses **Network Location Awareness (NLA)** to detect your network type and apply the correct firewall profile.

1.Private Network

- Your home or trusted network.
- Fewer restrictions → easier connections.

2.Guest/Public Network

- Untrusted networks (coffee shops, airports, restaurants).
- Recommended:
  - Allow only essential outgoing traffic
  - Block all incoming traffic

Allowing/Blocking Apps

- Click option **(1)** → list of apps appears.
- You can check/uncheck apps for **Private** or **Public** profiles.

Turning Firewall On/Off

- Click option **(2)**.
- Microsoft recommends *NOT* turning off the firewall → instead block incoming connections.

Restore Defaults

- Click option **(3)** to restore original firewall settings.

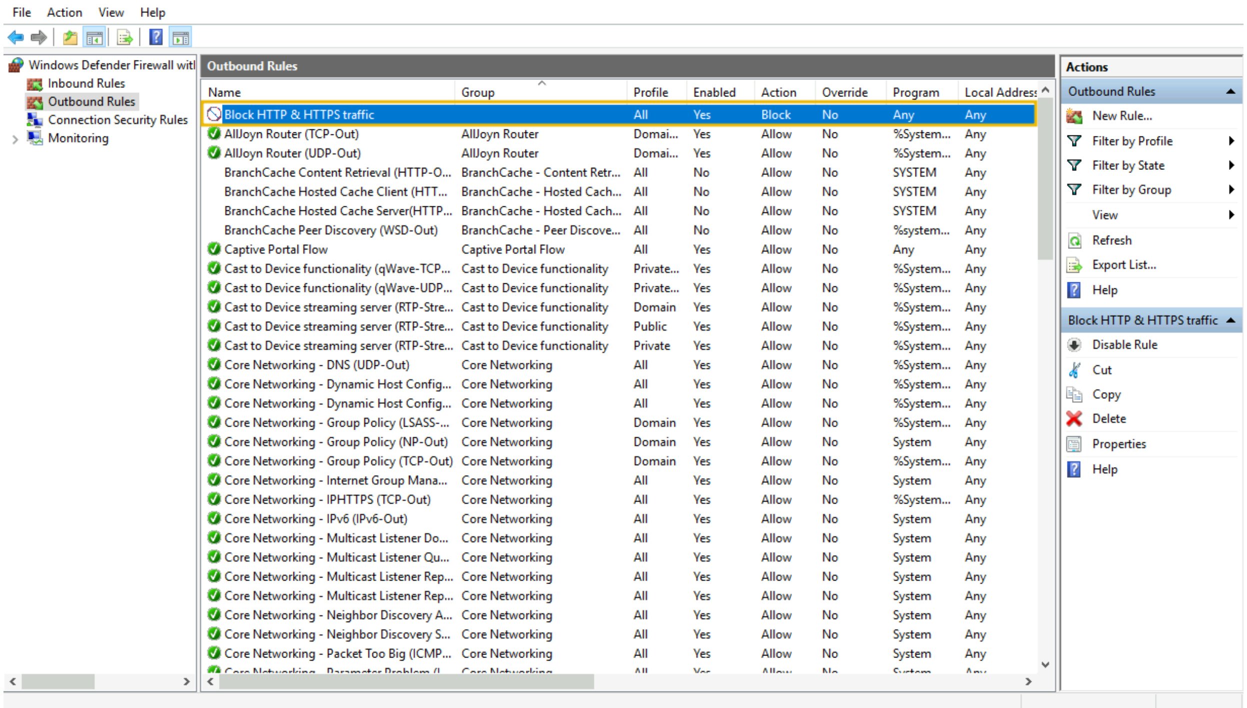
Custom Rules

You can create custom firewall rules to:

- Allow specific ports
- Block specific ports
- Restrict outgoing traffic
- Restrict incoming traffic

Example Task: Block Outgoing HTTP/HTTPS (Ports 80 & 443)

Goal:
Block browsing → Websites use ports **80 (HTTP)** and **443 (HTTPS)**.



Linux iptables Firewall

- **Linux** also provides **powerful built-in firewalls** to control network traffic.
- Linux's firewall system is built on top of a core framework called **Netfilter**.

What is Netfilter?

Netfilter is the **core firewall framework inside the Linux kernel**.
It handles:

- Packet filtering
- NAT (Network Address Translation)
- Connection tracking

Many Linux firewall tools use Netfilter in the background.

Common utilities that use Netfilter:

1. iptables

- Most widely used old firewall tool
- Works directly with Netfilter
- Very powerful, but syntax is complex

2. nftables

- Newer replacement for iptables
- Faster, simpler, more efficient
- Still uses Netfilter backend

3.firewalld

- Higher-level firewall tool
- Has pre-built **zones** (public, home, dmz, etc.)
- Easier to manage dynamic rules

UFW (Uncomplicated Firewall)

- **ufw** is the easiest firewall tool for Linux.
- It provides simple commands to manage complex iptables rules.

Basically:

You write simple **ufw commands → ufw writes complicated iptables rules for you.**

How to install ufw (for linux)

Sudo apt install ufw -y

Basic UFW Commands

1.Check Firewall Status

sudo ufw status

```
┌──(yuvi㉿kali)-[~]
└─$ sudo ufw status
Status: inactive
```

✔️ Enable UFW Firewall

sudo ufw enable

```
┌──(yuvi㉿kali)-[~]
└─$ sudo ufw enable
Firewall is active and enabled on system startup
```

✔️ Disable Firewall

sudo ufw disable

```
┌──(yuvi㉿kali)-[~]
└─$ sudo ufw disable
Firewall stopped and disabled on system startup
```

Allow All Outgoing Traffic (Default Policy)

This rule allows all outgoing connections unless blocked by a specific rule later.

sudo ufw default allow outgoing

```
┌──(yuvi㉿kali)-[~]
└─$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
```

You can do the same for **incoming** traffic by replacing outgoing with incoming.

❌ Deny Incoming SSH Traffic (Port 22)

To block SSH access:

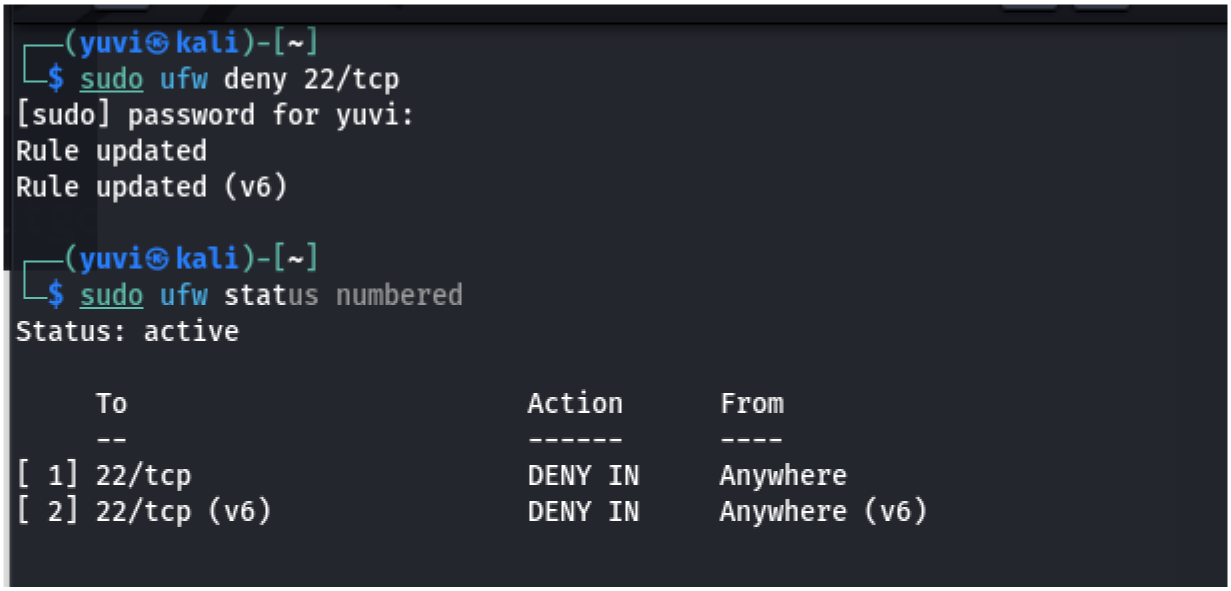
sudo ufw deny 22/tcp

```
┌──(yuvi㉿kali)-[~]
└─$ sudo ufw deny 22/tcp
[sudo] password for yuvi:
Rules updated
Rules updated (v6)
```

This blocks SSH for IPv4 and IPv6.

📋 List All Active Rules (Numbered)

sudo ufw status numbered



```
┌──(yuvi㉿kali)-[~]
└─$ sudo ufw deny 22/tcp
[sudo] password for yuvi:
Rule updated
Rule updated (v6)

┌──(yuvi㉿kali)-[~]
└─$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     DENY IN     Anywhere
[ 2] 22/tcp (v6)                DENY IN     Anywhere (v6)
```

Delete a Rule by Number

Example: deleting rule #2

sudo ufw delete 2

```
┌──(yuvi㉿kali)-[~]
└─$ sudo ufw delete 2
Deleting:
 deny 22/tcp
Proceed with operation (y|n)? y
Rule deleted (v6)

┌──(yuvi㉿kali)-[~]
└─$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     DENY IN     Anywhere
```

Choosing the Right Linux Firewall Tool

Use depends on:

- Your experience level
- How simple or advanced your firewall needs are
- Distribution defaults (**Ubuntu → UFW, CentOS/RHEL → firewalld**)

For beginners:
ufw is easiest

For advanced users:
iptables or **nftables**

For server admins:
Firewalld

A firewall is useful for a SOC Analyst because it:

- Blocks attacks
- Generates valuable logs
- Helps detect suspicious traffic
- Supports threat hunting
- Helps in incident response
- Enforces network segmentation
- Works with SIEM & TI systems

Reference

https://en.wikipedia.org/wiki/Firewall_(computing)

https://learn.microsoft.com/en-us/windows/security/operating-system-security/network-security/windows-firewall/rules

-yuvaraj.D